PLATFORM SHOWCASE · JULY 2026

CRIA

Colorado River Integrated Assessment

An observation-validated decision-support platform that brings the Colorado River Basin's surface water, snowpack and subsurface storage into one view — every number checked against NASA satellites and peer-reviewed science, in the units water managers already plan in.

Center for Hydrologic Innovations, Arizona State University · NASA Applied Sciences · in collaboration with the Central Arizona Project

The gap it fills

The Colorado River is over-allocated and drying, yet snowpack, storage, drought and climate projections still live in **separate tools and different units**, and surface and subsurface water are rarely shown together. CRIA gives the Technical Water Manager a single, consistent, observation-validated view of the whole basin — moving the project from a set of individual studies to an **integrated decision-support platform**.

What's new — the platform at a glance

01

Observation-validated key values

Every headline number is computed live from a PRISM-calibrated VIC 5.0 reanalysis and evaluated against NASA GRACE, SMAP and SNOTEL — not static figures typed into a slide.

02

Whole basin, one view

Surface water, snowpack and the water hidden underground (GRACE storage) are brought into a single, sub-basin picture — the surface and subsurface story, together.

03

Published★ validation markers

Key results carry a yellow-star badge that opens the peer-reviewed paper that reached the same result independently — GRACE (Castle 2014, Abdelmohsen 2025), Ghimire 2026, Wang 2026.

04

Research Team tab

A dedicated section presenting every contributor with verified professional profiles (LinkedIn / ASU) — who builds the platform and stands behind its numbers.

05

RIA assistant — with guardrails

A plain-language AI assistant answers questions on every tab. Confidential items — budgets, salaries, grant amounts, personal contact details — are **blocked by design**.

06

Reviewer questions, answered in-app

The hard questions a reviewer asks — is it real, is it circular, is the model trusted — are marked directly in the interface so the user reaches them easily.

07

Chat opens in its own page

When more room is needed, the assistant expands to a full page; it can also be minimised or reopened — the reading space adapts to the user.

08

Display-size control

Three text scales (reading, large, presentation) let the same platform serve a laptop review, a manager's desk, or a projected demo.

09

An engaging, curiosity-driven Overview

A live signal ticker, a dry-vs-wet October scene, and 3-D finding tiles draw the user in and invite them to explore rather than skim.

10

Decision-ready throughout

Supply and shortage in the units managers plan in — acre-feet and Lake Mead shortage tiers — with a sub-basin selector and live reservoir levels.

How it compares

A capability view of what an integrated basin decision-tool needs. Operational models such as the Bureau of Reclamation's CRSS remain authoritative for reservoir operations — CRIA complements them by providing the observation-validated hydrologic diagnosis behind the numbers.

Capability	Data portals (e.g. CDSS)	Operations model (e.g. CRSS)	Scenario Explorer	CRIA
Surface water balance	◓	◓	✓	✓
Snowpack → runoff	◓	—	◓	✓
Subsurface / groundwater storage (GRACE)	—	—	—	✓
Multi-sensor validation shown in-app	—	—	◓	✓
Sub-basin resolution	✓	✓	◓	✓
Decision units & reservoir tiers	◓	✓	—	✓
Uncertainty stated & provenance	◓	◓	◓	✓
Explainability (guided + assistant)	—	—	—	✓

✓ full · ◓ partial · — not covered

Comparison is capability-based and respectful of each tool's purpose; it is not a performance ranking.

Validation & standing in the literature

Three findings, each carrying the statistic behind it and the peer-reviewed study that reached it independently:

$p = 10^{-19}$

The basin's water loss is real — and largely underground

GRACE terrestrial-storage trend, independent of the model

★ Castle et al. 2014; Abdelmohsen et al. 2025 (GRL)

LOO $R^2 = 0.45$

October soil moisture forecasts the water year

Leave-one-out cross-validation over 41 water years

★ Ghimire, Vivoni & Wang 2026 (Water Resources Research)

NSE = 0.96

The model reproduces observed streamflow

Upper-Basin gauges; soil moisture validated vs NASA SMAP (R^2 0.81)

★ Wang et al. 2026 (Scientific Reports)

OPEN QUESTIONS THE PLATFORM RAISES — NO STUDY HAS ANSWERED THEM YET

1. Does October soil moisture add skill beyond the operational 24-Month Study?
2. Could a live, standing 1-October basin index be operationalised?
3. Can autumn storage be managed as a lever on next-year supply?

Model skill: NSE ≈ 0.96 (Upper-Basin streamflow) · SMAP soil-moisture R^2 0.71–0.81 · GRACE terrestrial-storage evaluated. Data: PRISM-calibrated VIC 5.0 reanalysis (WY1982–2024), NASA GRACE/GRACE-FO, SMAP L4, SNOTEL (103 stations), USBR shortage tiers.

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